Year 13 Maths - Rational functions and partial fractions

Similar questions - Partial fractions in calculus and modelling

Two new questions on this topic, a little more open-ended than the exam questions. They are not past exam questions. Answers are at the end.

1 A curve satisfies the differential equation $\dfrac{\mathrm{d}y}{\mathrm{d}x} = \dfrac{3y}{(x-1)(x+2)}$ for $x > 1$, and passes through the point $(2,2)$.

(a) Express $\dfrac{3}{(x-1)(x+2)}$ in partial fractions. **[3]**

(b) Solve the differential equation, giving y in terms of x. **[6]**

(c) Describe what happens to y as $x \to \infty$. **[1]**

2 The size P of a population at time t years is modelled by $\dfrac{\mathrm{d}P}{\mathrm{d}t} = 0.002P(400-P)$, where $P = 100$ when $t = 0$.

(a) Show that $\dfrac{1}{P(400-P)} = \dfrac{1}{400}\left(\dfrac{1}{P} + \dfrac{1}{400-P}\right)$. **[2]**

(b) Solve the differential equation to show that $P = \dfrac{400}{1+3\mathrm{e}^{-0.8t}}$. **[7]**

(c) Find the time taken for the population to reach 300, and state the long-term population predicted by the model. **[3]**

Question 1

(a) $\dfrac{3}{(x-1)(x+2)} = \dfrac{A}{x-1} + \dfrac{B}{x+2}$: $3 = A(x+2) + B(x-1)$. $x = 1$: $A = 1$; $x = -2$: $B = -1$. So $\dfrac{1}{x-1} - \dfrac{1}{x+2}$

(b) $\displaystyle\int \frac{1}{y}\,\mathrm{d}y = \int \left(\frac{1}{x-1} - \frac{1}{x+2}\right)\mathrm{d}x$, so $\ln y = \ln(x-1) - \ln(x+2) + c$ and $y = A\,\dfrac{x-1}{x+2}$. At $(2,2)$: $2 = A \times \frac{1}{4}$, so $A = 8$.

$y = \dfrac{8(x-1)}{x+2}$

(c) $y = 8\,\dfrac{1-1/x}{1+2/x} \to 8$ as $x \to \infty$

Question 2

(a) $\dfrac{1}{P} + \dfrac{1}{400-P} = \dfrac{400-P+P}{P(400-P)} = \dfrac{400}{P(400-P)}$, so dividing by 400 gives the result

(b) $\displaystyle\int \frac{1}{P(400-P)}\,\mathrm{d}P = \int 0.002\,\mathrm{d}t$ gives $\dfrac{1}{400}\ln\dfrac{P}{400-P} = 0.002t + c$, so $\ln\dfrac{P}{400-P} = 0.8t + c'$. At $t = 0$: $\ln\frac{1}{3} = c'$.

So $\dfrac{P}{400-P} = \frac{1}{3}\mathrm{e}^{0.8t}$, giving $3P = (400-P)\mathrm{e}^{0.8t}$ and $P = \dfrac{400}{1+3\mathrm{e}^{-0.8t}}$

(c) $300 = \dfrac{400}{1+3\mathrm{e}^{-0.8t}}$ gives $\mathrm{e}^{-0.8t} = \frac{1}{9}$, so $t = \dfrac{\ln 9}{0.8} = 2.75$ years. As $t \to \infty$, $P \to 400$